

Adaptive exact minimax rates for invariant density estimation on quotient spaces

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Abstract

We study nonparametric density estimation when the target is invariant under a compact group G acting on the sample space E . The orbit projection is sufficient, so the problem reduces *exactly* to density estimation on the quotient $M = E/G$ of effective dimension $d = \dim(E/G) = \dim E - \dim(\text{orbit})$. We show that the minimax squared-Hellinger risk over the invariant Hölder class $\Sigma_{\text{inv}}(s, L)$ is of order $n^{-2s/(2s+d)}$, with matching constants up to the rate: a Fano lower bound built from G -invariant bumps on the quotient, paired with two adaptive upper bounds. An orbital minimum-description-length (MDL) estimator attains the rate up to a logarithmic factor; we prove this factor is a *coding artefact* and remove it with a penalised projection estimator that attains the exact rate $n^{-2s/(2s+d)}$ adaptively in the unknown smoothness s . The structural reason is isolated: a projection estimator pays the model’s metric complexity D_k/n , whereas a description-length code additionally pays the per-coefficient quantisation cost $\log(1/\delta) \asymp \log n$. We then extend the results to a *multiplicative* (logarithmic-twin) regime—positive or compositional data—where the Hellinger-affinity oracle step is tail-free, so the gained rate persists on heavy-tailed data for which identity-coordinate concentration is unavailable. We position the dimension-reduction phenomenon against the recent “exact sample-complexity gain from invariance” literature and delineate precisely what is a re-derivation and what is new.

Keywords. Invariant density estimation; minimax rates; group action; quotient space; effective dimension; minimum description length; penalised projection; adaptive estimation; Hellinger affinity.

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1 Introduction

1.1 Problem and main result

Let a compact group G act on a sample space E , and suppose the unknown density $f^\star$ of the i.i.d. data $X_1, \dots, X_n$ is G -invariant: $f^\star(g \cdot x) = f^\star(x)$ for all $g \in G$. Such structural priors are ubiquitous: radial (rotation-invariant) densities, partially translation-invariant fields, exchangeable or permutation-invariant models, and compositional data with scale or sub-composition symmetries. The statistical question is how much the invariance is worth.

A G -invariant density descends to a density $\tilde{f}^\star$ on the quotient $M := E/G$, a metric space of *effective dimension*

$$d := \dim(E/G) = \dim E - \dim(\text{generic orbit}) = D - r, \quad (1)$$

where $D = \dim E$ and r is the dimension of a principal orbit. When G acts with *positive-dimensional* orbits ($r > 0$) the quotient is genuinely lower-dimensional, and one expects the estimation rate to be governed by d rather than by D . Our main result makes this exact and adaptive.

Theorem 1.1 (Exact adaptive minimax rate; informal). *Under the standing assumptions of Section 2, the minimax squared Hellinger risk over the invariant Hölder ball $\Sigma_{\text{inv}}(s, L)$ satisfies*

$$\inf_{\hat{f}} \sup_{f^\star \in \Sigma_{\text{inv}}(s, L)} \mathbb{E} h^2(P^\star, \hat{f}) \asymp n^{-2s/(2s+d)}, \quad d = \dim(E/G).$$

The lower bound holds for all $s > 0$; the upper bound is attained, for the exact rate and without knowledge of s , by a penalised projection estimator on the quotient. The invariant rate strictly improves on the non-invariant rate $n^{-2s/(2s+D)}$ if and only if the orbits are positive-dimensional.

Two estimators appear. An *orbital MDL* estimator—selecting a G -invariant candidate density by minimising negative log-likelihood plus a Kraft code length—attains the rate $(\log n/n)^{2s/(2s+d)}$, i.e. the minimax rate up to a logarithmic factor. We then show that this $\log n$ is not intrinsic: it is the per-coefficient cost of quantising a continuum into a prefix code. A penalised projection estimator, which fits the coefficients without quantising them, pays only the metric complexity of the model and attains the *exact* rate, adaptively.

1.2 Contributions

- (1) **A clean reduction and a complete lower bound.** We prove that the orbit projection $\pi(X_i)$ is sufficient and that the invariant minimax risk equals the minimax risk on the quotient (Lemma 2.1), and we give a self-contained Fano lower bound $n^{-2s/(2s+d)}$ using a Varshamov–Gilbert cube of G -invariant bumps supported on the quotient (Theorem 3.4).

- (2) **An information-theoretic (MDL/Hellinger) upper bound.** A tail-free Kraft/partition-function bound (Lemma 4.1) yields a Hellinger oracle inequality with *no* tail or moment assumption; combined with a twin Jackson estimate on the (possibly stratified) quotient it gives the adaptive rate $(\log n/n)^{2s/(2s+d)}$ (Theorem 4.6).
- (3) **Removal of the logarithmic factor.** We prove that a single-resolution projection estimator attains $n^{-2s/(2s+d)}$ exactly for known s (Proposition 5.1) and that a penalised projection estimator attains it *adaptively* (Theorem 5.2), and we isolate the mechanism: metric complexity D_k/n versus the coding cost $\log(1/\delta)$ of the description-length penalty (Remark 5.3).
- (4) **A multiplicative / twin extension.** On positive or compositional data in a logarithmic chart, the affinity oracle step is tail-free and the loss is a change-of-variables invariant, so the exact rate persists on heavy-tailed data where identity-coordinate concentration fails (Section 7).
- (5) **Honest positioning.** We delineate which parts re-derive the known “invariance lowers the effective dimension” phenomenon and which are new (Section 1.3).

1.3 Related work and novelty

That invariance reduces the effective dimension of a learning problem, and that the gain is governed by the dimension of the quotient, is by now well understood, particularly in the L^2 /Sobolev and kernel settings. Tahmasebi and Jegelka [10] quantify the *exact* sample-complexity gain from invariances for kernel ridge regression on manifolds; Elesedy and Zaidi [5] and Bietti, Venturi and Bruna [3] give related generalisation and approximation analyses for invariant/equivariant models. Classical nonparametric density estimation on manifolds and quotient spaces (Hendriks [6], Pelletier [9]) already produces rates governed by the intrinsic dimension. Against this background, the bare statement of Theorem 1.1 in the L^2 /Sobolev setting is a re-derivation and we present it as such.

What we add is threefold. First, an *information-theoretic* route: the rate emerges from the Barron–Cover [1] Kraft partition-function bound in Hellinger risk, rather than from RKHS spectra, and the resulting estimator is adaptive by construction. Second, an explicit and rigorous account of the *logarithmic gap* between description-length coding (near-minimax) and projection/penalised estimation (exactly minimax), via the Birgé–Massart theory [4, 8]; this clarifies a point that is often elided when MDL or sieve-MDL procedures are called “minimax optimal”. Third, the *multiplicative / twin* regime (Section 7): the invariance is taken with respect to a group acting in a logarithmic (non-additive) chart and smoothness is measured there, so the gain extends to positive/compositional and heavy-tailed data, a regime outside the cited additive analyses. The tail-freeness of the affinity bound (Lemma 4.1) is what makes this extension automatic.

1.4 Notation and organisation

We write $h^2(P, Q) = \frac{1}{2} \int (\sqrt{p} - \sqrt{q})^2 d\mu$ for the squared Hellinger distance, $H = \sqrt{h^2}$ for the Hellinger metric, $\rho(f, g) = \int \sqrt{fg} d\mu \in (0, 1]$ for the affinity (so $h^2 = 1 - \rho$), and KL for Kullback–Leibler divergence. Throughout $M = E/G$, $d = \dim M$, $\pi : E \rightarrow M$ is the quotient map, $\mu_G = \pi_* \mu$ the quotient reference measure, and $\Sigma(s, L)$ the Hölder ball of order s and radius L on M , with $\Sigma_{\text{inv}}(s, L) = \{\tilde{f} \circ \pi : \tilde{f} \in \Sigma(s, L)\}$ its invariant pullback. Section 2 sets up the reduction; Section 3 proves the lower bound; Section 4 the MDL near-minimax upper bound; Section 5 the exact-rate projection estimators; Section 6 assembles the matching rate and the examples; Section 7 the multiplicative extension; Section 8 gives a numerical illustration; Section 9 discusses the stratified case and open points.

2 Setting and reduction to the quotient

Assumption 1 (Geometry and group). (E, ρ_E) is a compact metric space (typically a compact Riemannian manifold of dimension D). A compact group G acts continuously on E by isometries with closed orbits; principal orbits have dimension r . The quotient $M = E/G$ carries the *infimum (orbital) metric*

$$d_G([x], [y]) := \inf_{g \in G} \rho_E(x, g \cdot y),$$

and has covering dimension $d = \dim M = D - r$ for a smooth proper action.

Assumption 2 (Reference measure and invariant class). Fix a G -invariant base measure μ on E with pushforward $\mu_G = \pi_* \mu$. The target is G -invariant, $f^* = \tilde{f}^* \circ \pi$, with $\tilde{f}^* \in \Sigma(s, L)$ and $0 < c_0 \leq \tilde{f}^* \leq C_1 < \infty$ on a region $U \subseteq M$ on which (M, d_G, μ_G) is Ahlfors d -regular: $c_M^{-1} r^d \leq \mu_G(B(x, r)) \leq c_M r^d$ for $x \in U$, $0 < r \leq r_0$.

Assumption 3 (Stratified quotient with full-mass top stratum). The quotient decomposes as $M = M_0 \sqcup S$, where the *principal stratum* M_0 is an open d -dimensional manifold on which (M_0, d_G, μ_G) is Ahlfors d -regular (Assumption 2, with $U \subseteq M_0$), and the singular locus S (non-principal orbit types) satisfies $\dim S < d$, hence $\mu_G(S) = 0$. For a free action $S = \emptyset$ and M is a compact d -manifold.

Assumption 3 is what makes the orbifold case rigorous: by Ahlfors d -regularity any set of dimension $< d$ is μ_G -null, so all $L^2(\mu_G)$ - and KL-integrals—approximation, projection variance, and the information radius of the lower bound—are computed on the regular top stratum M_0 , a genuine d -dimensional space. The only case it excludes is a target whose smoothness interacts essentially with the singular strata (where the local dimension drops); we return to this in Section 9. Throughout, “on M ” may be read “on M_0 ”.

The invariance prior in Assumption 2 is what makes the gain real: without it, orbit projection estimates only the invariant part $\pi_* P^*$ and there is nothing to gain. The following lemma is the formal reduction.

Lemma 2.1 (Sufficiency of the orbit projection). *Let $\mathcal{F}_{\text{inv}}$ be a family of G -invariant densities on E and $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} P_f$, $f \in \mathcal{F}_{\text{inv}}$. Then $(\pi(X_1), \dots, \pi(X_n))$ is sufficient for f , is i.i.d. from $\tilde{P}_{\tilde{f}} = \pi_* P_f$ on (M, μ_G) , and $h_E^2(P_f, P_g) = h_M^2(\tilde{P}_{\tilde{f}}, \tilde{P}_{\tilde{g}})$. Consequently*

$$\inf_{\tilde{f}} \sup_{f \in \mathcal{F}_{\text{inv}}} \mathbb{E} h_E^2(P_f, \hat{f}) = \inf_{\tilde{f}} \sup_{\tilde{f}} \mathbb{E} h_M^2(\tilde{P}_{\tilde{f}}, \hat{\tilde{f}}).$$

Proof. Invariance $f = \tilde{f} \circ \pi$ gives $f(X_i) = \tilde{f}(\pi(X_i))$, so the likelihood factors through $\pi(X_i)$: this is sufficiency, and the pushforward of an i.i.d. sample is i.i.d. For the Hellinger identity, invariant densities make the affinity integrate to the quotient, $\int_E \sqrt{fg} d\mu = \int_M \sqrt{\tilde{f}\tilde{g}} d\mu_G$. For the equality of minimax risks, note that under any P_f with f invariant, the conditional law of X_i given $\pi(X_i) = o$ is the normalised G -invariant measure on the orbit $\pi^{-1}(o)$, which does not depend on f : the within-orbit position is *ancillary* with f -free law. Hence the experiment $(X_{1:n}; \{P_f\})$ is Le Cam-equivalent to $(\pi(X_{1:n}); \{\tilde{P}_{\tilde{f}}\})$ augmented by independent f -free randomisation, and the two minimax risks coincide; squared Hellinger, being invariant, is the same loss on both sides. $\square$

Lemma 2.1 reduces every statement below to density estimation of $\tilde{f}^*$ on the d -dimensional space (M, μ_G) from n i.i.d. samples. We henceforth work on M and write f^* for $\tilde{f}^*$ when no confusion arises.

3 Lower bound

We construct a Varshamov–Gilbert cube of G -invariant perturbations supported on the quotient and apply Fano’s method.

Fix a $\mathcal{C}^\infty$ function $\psi : \mathbb{R}^d \rightarrow \mathbb{R}$ with $\text{supp } \psi \subseteq (0, 1)^d$, $\int \psi = 0$, $\|\psi\|_\infty \leq 1$, $\|\psi\|_{\mathcal{C}^s} \leq 1$, $\|\psi\|_{L^2} > 0$. Identifying U with a Euclidean cube and μ_G with Lebesgue measure up to c_M , for a bandwidth $h = 1/m$ split U into $N \asymp c_M h^{-d}$ sub-cubes $\{U_j\}$ of side h and centres x_j , and set

$$\psi_j(x) := L h^s \psi\left(\frac{x-x_j}{h}\right), \quad \tilde{f}_\omega := f_0 + \sum_{j=1}^N \omega_j \psi_j, \quad \omega \in \{0, 1\}^N, \quad (2)$$

where $f_0 \in \Sigma(s, L/2)$ is a fixed reference with $c_0 \leq f_0 \leq C_1$ on U .

Lemma 3.1 (Admissibility). *For $h \leq h_0$ (so $Lh^s \leq c_0/2$), each $\tilde{f}_\omega$ is a probability density and $\tilde{f}_\omega \in \Sigma(s, c_1 L)$ for a constant $c_1 = c_1(\psi)$.*

Proof. $\int \psi_j = Lh^s h^d \int \psi = 0$, so $\int \tilde{f}_\omega = 1$; and $|\sum_j \omega_j \psi_j| \leq Lh^s \leq c_0/2$, so $\tilde{f}_\omega \geq c_0/2 > 0$. The bumps have disjoint supports and $\|\psi_j\|_{\mathcal{C}^s} \leq L$; for disjoint supports the $\mathcal{C}^s$ seminorm of the sum is the maximum over j , whence $\|\tilde{f}_\omega\|_{\mathcal{C}^s} \leq \|f_0\|_{\mathcal{C}^s} + L \leq c_1 L$. $\square$

Lemma 3.2 (Varshamov–Gilbert [11, Lem. 2.9]). *For $N \geq 8$ there is $\Omega \subseteq \{0, 1\}^N$ with $|\Omega| \geq 2^{N/8}$ and $\text{Ham}(\omega, \omega') \geq N/8$ for distinct $\omega, \omega' \in \Omega$. Fix such an Ω and a base point $\omega^{(0)} \in \Omega$.*

Lemma 3.3 (Separation and information radius). *For distinct $\omega, \omega' \in \Omega$,*

$$h^2(\tilde{f}_\omega, \tilde{f}_{\omega'}) \geq c_2 L^2 h^{2s}, \quad c_2 = \frac{c_M \|\psi\|_{L^2}^2}{64 C_1},$$

and, with $P_\omega^{\otimes n}$ the sample law, $\text{KL}(P_\omega^{\otimes n} \| P_{\omega^{(0)}}^{\otimes n}) = n \text{KL}(\tilde{f}_\omega \| \tilde{f}_{\omega^{(0)}}) \leq c_3 n L^2 h^{2s}$.

Proof. $(\sqrt{a} - \sqrt{b})^2 \geq (a - b)/(4C_1)$ since $\tilde{f}_\omega, \tilde{f}_{\omega'} \in [c_0/2, 2C_1]$. Disjoint supports give $\int (\tilde{f}_\omega - \tilde{f}_{\omega'})^2 = \text{Ham}(\omega, \omega') L^2 h^{2s+d} \|\psi\|_{L^2}^2$, so $h^2 \geq \frac{1}{8C_1} \text{Ham} \cdot L^2 h^{2s+d} \|\psi\|_{L^2}^2 \geq c_2 L^2 h^{2s}$ using $\text{Ham} \geq N/8 \asymp c_M h^{-d}/8$. For the divergence, $\text{KL} \leq \chi^2 \leq \frac{2}{c_0} \int (\tilde{f}_\omega - \tilde{f}_{\omega^{(0)}})^2 \leq \frac{2}{c_0} N L^2 h^{2s+d} \|\psi\|_{L^2}^2 \leq c_3 L^2 h^{2s}$ (per observation), then tensorise. $\square$

Theorem 3.4 (Minimax lower bound on the quotient). *Under Assumptions 1–3 there are $c, n_0 > 0$ with*

$$\inf_{\hat{f}} \sup_{f^* \in \Sigma_{\text{inv}}(s, L)} \mathbb{E} h^2(P^*, \hat{f}) \geq c n^{-2s/(2s+d)}, \quad n \geq n_0.$$

Proof. By Lemma 2.1 it suffices to prove the bound on (M, μ_G) . Take the family $\{\tilde{f}_\omega : \omega \in \Omega\}$ of Lemmas 3.1–3.2 (all in $\Sigma(s, c_1 L)$, hence in $\Sigma(s, L)$ after rescaling L). Choose $h = h_n$ by $n L^2 h_n^{2s+d} = c_4$, i.e. $h_n = (c_4/(n L^2))^{1/(2s+d)}$; then $N \asymp c_M h_n^{-d} \rightarrow \infty$ so $h_n \leq h_0$ and $N \geq 8$ for $n \geq n_0$. By Lemma 3.3, $H(\tilde{f}_\omega, \tilde{f}_{\omega'}) \geq \sqrt{c_2} L h_n^s =: 2\delta_n$, and

$$\frac{1}{|\Omega|} \sum_{\omega} \text{KL}(P_\omega^{\otimes n} \| P_{\omega^{(0)}}^{\otimes n}) \leq c_3 n L^2 h_n^{2s} = c_3 c_4 h_n^{-d}, \quad \log |\Omega| \geq \frac{N}{8} \log 2 \geq c_5 h_n^{-d},$$

so the Fano ratio is $\leq c_3 c_4 / c_5 =: \alpha$, which is $< 1/8$ for c_4 small. By the Fano method [11, Thm. 2.7], $\inf_{\hat{f}} \max_{\omega} P_\omega(H(\hat{f}, \tilde{f}_\omega) \geq \delta_n) \geq c_* > 0$, and Markov’s inequality gives $\inf_{\hat{f}} \sup_{\Sigma} \mathbb{E} h^2 \geq \delta_n^2 c_* = \frac{c_2 c_*}{4} L^2 h_n^{2s} \asymp n^{-2s/(2s+d)}$ (the L -dependence absorbed into c). Pulling back to E via Lemma 2.1 completes the proof. $\square$

4 Upper bound by orbital MDL (near-minimax)

Let $\{f_m\}_{m \in \mathcal{I}}$ be a countable family of G -invariant candidate densities with prefix code lengths $L(m)$ satisfying the half-Kraft condition

$$\sum_{m \in \mathcal{I}} 2^{-L(m)/2} \leq 1 \quad (3)$$

(obtained from any Kraft code by $L \leftarrow 2L$). The orbital MDL estimator is

$$\hat{m} \in \arg \min_m \left\{ - \sum_{i=1}^n \log f_m(X_i) + L(m) \log 2 \right\}, \quad \hat{f} = f_{\hat{m}}. \quad (4)$$

Lemma 4.1 (Tail-free Kraft/partition-function bound). *For the estimator (4) under (3),*

$$\mathbb{E}_{P^*} \left[2^{-L(\hat{m})/2} \prod_{i=1}^n \sqrt{\frac{f_{\hat{m}}(X_i)}{f^*(X_i)}} \right] \leq \sum_m 2^{-L(m)/2} \rho(f^*, f_m)^n \leq 1.$$

No tail or moment assumption on f^ or f_m is used.*

Proof. The selector (4) maximises $2^{-L(m)} \prod_i f_m(X_i)$ over m ; taking square roots, for every realisation $2^{-L(\hat{m})/2} \prod_i \sqrt{f_{\hat{m}}(X_i)} = \max_m 2^{-L(m)/2} \prod_i \sqrt{f_m(X_i)} \leq \sum_m 2^{-L(m)/2} \prod_i \sqrt{f_m(X_i)}$. Divide by $\prod_i \sqrt{f^*(X_i)} > 0$, take $\mathbb{E}_{P^*}$, and use $\mathbb{E}_{P^*} \sqrt{f_m/f^*} = \rho(f^*, f_m)$ and independence; finally $\rho \leq 1$ and (3). $\square$

Theorem 4.2 (Hellinger oracle inequality). *There is a universal C with $\mathbb{E}_{P^*}[h^2(P^*, \hat{f})] \leq C \inf_{m \in \mathcal{I}} \{\text{KL}(P^* \| f_m) + (\log 2)L(m)/n\}$.*

Proof. Lemma 4.1 is the index-of-resolvability estimate of Barron–Cover [1] (see also Li [7]); the passage to the displayed bound combines a Markov step on the random affinity with $h^2 \leq -\log \rho \leq \text{KL}$. $\square$

Remark 4.3 (Tails are irrelevant to concentration). Lemma 4.1 uses only $\rho \leq 1$, valid for any pair of densities; the concentration step carries *no* tail hypothesis. Tails can enter only through the approximation term $\text{KL}(P^* \| f_m)$, i.e. through smoothness. This is what powers the multiplicative extension of Section 7.

Lemma 4.4 (Orbital sieve and twin Jackson bound). *Under Assumptions 2–3 and $\tilde{f}^* \in \Sigma(s, L)$, for each $k \in \mathbb{N}$ there is a G -invariant density q_k , piecewise polynomial of degree $< \lceil s \rceil$ on a resolution- 2^{-k} partition of U pulled back through π , with $\dim \mathcal{M}_k \asymp 2^{kd}$, $c_0/2 \leq q_k \leq 2C_1$, $\|\tilde{f}^* - q_k\|_\infty \leq CL2^{-ks}$, hence $\text{KL}(P^* \| Q_k) \leq C' L^2 2^{-2ks}$.*

Proof. By d -Ahlfors regularity any subset of dimension $< d$ is μ_G -null; in particular the non-principal strata carry no mass, so the $L^2(\mu_G)$ and KL approximation errors are computed on the principal stratum, a d -dimensional Ahlfors-regular space, where the classical piecewise-polynomial Jackson estimate gives $\|\tilde{f}^* - q_k\|_\infty \leq CL2^{-ks}$ on each cell. Truncating to $[c_0/2, 2C_1]$ and renormalising changes the sup-norm by $O(2^{-ks})$; finally $\text{KL} \leq \chi^2 \leq \frac{2}{c_0} \|\cdot\|_{L^2}^2 \leq C' L^2 2^{-2ks}$. $\square$

Lemma 4.5 (Code length). *Quantising the $\asymp 2^{kd}$ coefficients of $\mathcal{M}_k$ on a grid of spacing $\delta_k \asymp 2^{-ks}$ yields a countable family with a prefix code satisfying (3) and $L(k) \asymp 2^{kd} \log_2(1/\delta_k) \lesssim 2^{kd} \log n$ at the resolution used below, while the quantisation adds at most $C2^{kd}/n$ to $\text{KL}(P^* \| q_k)$.*

Proof. Standard sieve discretisation [1]: a grid of spacing δ_k on each of $\asymp 2^{kd}$ bounded coefficients costs $\log_2(1/\delta_k)$ bits per coefficient, and an Elias code on k adds $O(\log k)$ bits. The quantisation perturbs q_k in sup-norm by $O(\delta_k)$, hence KL by $O(\delta_k^2 2^{kd})$; with $\delta_k \asymp 2^{-ks}$ at the balance $2^k \asymp (n/\log n)^{1/(2s+d)}$ this is $O(2^{kd}/n)$. $\square$

Theorem 4.6 (Adaptive orbital MDL upper bound). *Under Assumptions 1–3, the orbital MDL estimator (4) over the family of Lemmas 4.4–4.5 satisfies, for every $s \in (0, s_{\max}]$ and $f^* \in \Sigma_{\text{inv}}(s, L)$,*

$$\mathbb{E}_{P^*}[h^2(P^*, \hat{f})] \leq C \left(\frac{\log n}{n} \right)^{2s/(2s+d)},$$

adaptively (the estimator does not use s).

Proof. Combine Theorem 4.2 with Lemmas 4.4–4.5: $\mathbb{E}[h^2] \leq C \inf_k \{L^2 2^{-2ks} + 2^{kd} \log n/n\}$, and balancing $2^{k(2s+d)} \asymp n/\log n$ gives the rate. The infimum over k is realised by the data-driven $\hat{m}$, hence adaptivity. $\square$

5 The exact rate: projection estimators

The $(\log n)^{2s/(2s+d)}$ surplus of Theorem 4.6 is an artefact of coding: Lemma 4.5 spends $\log_2(1/\delta_k) \asymp \log n$ bits to quantise each of the $\asymp 2^{kd}$ coefficients. We remove it.

Let $\mathcal{M}_k$ be the linear space of G -invariant functions piecewise polynomial of degree $< \lceil s \rceil$ on the resolution- 2^{-k} partition of Lemma 4.4 (dimension $D_k \asymp 2^{kd}$), $\Pi_k : L^2(\mu_G) \rightarrow \mathcal{M}_k$ the orthogonal projection, and

$$\hat{g}_k := \Pi_k \left(\frac{1}{n} \sum_{i=1}^n \delta_{\pi(X_i)} \right), \quad \hat{f}_k := \frac{T(\hat{g}_k)}{\int T(\hat{g}_k) d\mu_G}, \quad T(u) = (u \vee \frac{c_0}{2}) \wedge 2C_1,$$

so $\hat{f}_k$ is a density in $[c_0/2, 2C_1]$ and T is a 1-Lipschitz contraction onto an interval containing f^* .

Proposition 5.1 (Known s : exact rate). *Fix s and let $2^{k_n} \asymp n^{1/(2s+d)}$. Then for every $f^* \in \Sigma_{\text{inv}}(s, L)$,*

$$\mathbb{E}_{P^*}[h^2(P^*, \hat{f}_{k_n})] \leq C(2^{-2k_n s} + D_{k_n}/n) \leq C' n^{-2s/(2s+d)},$$

with no logarithmic factor.

Proof. Since $\hat{f}_k \in [c_0/2, 2C_1]$ and $f^* \in [c_0, C_1]$, $h^2 \leq \frac{1}{c_0} \|\hat{f}_k - f^*\|_2^2$, and T with renormalisation only decreases the L^2 distance to f^* up to a universal factor; so it suffices to bound $\mathbb{E}\|\hat{g}_k - f^*\|_2^2$. Orthogonality splits it into bias and variance. Bias: the projection beats the interpolant q_k of Lemma 4.4, so $\|f^* - \Pi_k f^*\|_2 \leq CL 2^{-k_n s}$. Variance: with (e_ℓ) an $L^2(\mu_G)$ -orthonormal basis of $\mathcal{M}_k$, $\hat{g}_k - \Pi_k f^* = \sum_\ell (\frac{1}{n} \sum_i e_\ell(\pi X_i) - \mathbb{E} e_\ell) e_\ell$ and

$$\mathbb{E}\|\hat{g}_k - \Pi_k f^*\|_2^2 = \frac{1}{n} \sum_{\ell=1}^{D_k} \text{Var}(e_\ell(\pi X)) \leq \frac{1}{n} \int (\sum_\ell e_\ell^2) f^* d\mu_G \leq \frac{C_1}{n} \sum_\ell \|e_\ell\|_2^2 = \frac{C_1 D_k}{n}.$$

Balancing $2^{-2k_n s} \asymp D_{k_n}/n$ gives $C' n^{-2s/(2s+d)}$. $\square$

Theorem 5.2 (Adaptive exact rate via penalised projection). *Let $\hat{k} \in \arg \min_k \{-\|\hat{g}_k\|_2^2 + \kappa D_k/n\}$ for a large enough universal κ , and $\hat{f} = \hat{f}_{\hat{k}}$. Then for every $s \in (0, s_{\max}]$ and $f^* \in \Sigma_{\text{inv}}(s, L)$,*

$$\mathbb{E}_{P^*}[h^2(P^*, \hat{f})] \leq C \inf_k \left\{ \|f^* - \Pi_k f^*\|_2^2 + \frac{D_k}{n} \right\} \leq C' n^{-2s/(2s+d)},$$

adaptively in s , with no logarithmic factor.

Proof. The criterion is the least-squares contrast $\gamma_n(g) = \|g\|_2^2 - \frac{2}{n} \sum_i g(\pi X_i)$, minimised over $\mathcal{M}_k$ by $\hat{g}_k$ with $\gamma_n(\hat{g}_k) = -\|\hat{g}_k\|_2^2$; thus $\hat{k}$ is the penalised model-selection rule. The collection $\{\mathcal{M}_k\}$ is nested with geometrically growing dimensions $D_k \asymp 2^{kd}$. The Birgé–Massart penalised-projection oracle inequality [4, 8] then gives, whenever the weights $\{x_k\}$ satisfy $\Sigma := \sum_k e^{-x_k} < \infty$,

$$\mathbb{E} \|\hat{g}_{\hat{k}} - f^*\|_2^2 \leq C \inf_k \left\{ \|f^* - \Pi_k f^*\|_2^2 + \frac{D_k + x_k}{n} \right\} + \frac{C\Sigma}{n}.$$

Because $D_k \asymp 2^{kd}$ grows geometrically, $x_k = D_k$ already gives $\Sigma = \sum_k e^{-D_k} < \infty$ with $x_k \asymp D_k$, so the adaptation term x_k/n is of the same order D_k/n as the variance and contributes no extra factor; hence $\text{pen}(k) = \kappa D_k/n$ is admissible. Passing to $\hat{f}$ as in Proposition 5.1 and inserting the Jackson bias $\|f^* - \Pi_k f^*\|_2 \leq CL2^{-ks}$, the infimum is the balance of Proposition 5.1, now realised without knowing s . $\square$

Remark 5.3 (The mechanism: metric complexity versus coding cost). The contrast is structural. A description-length code (Lemma 4.5) discretises the continuum and pays $\log(1/\delta_k) \asymp \log n$ bits per coefficient, so its per-model penalty is $L(k) \asymp D_k \log n$ and its rate carries a log. A projection / penalised-projection estimator never quantises the coefficients; it pays only the metric complexity D_k/n . The geometric growth of D_k keeps the model-selection weights summable with linear-in- D_k exponents, so no log re-enters. The MDL estimator is therefore *near-minimax* and the projection estimator *exactly* minimax.

6 Matching rate, the gain from invariance, and examples

Theorem 6.1 (Exact adaptive minimax rate). *Under Assumptions 1–3, the minimax squared Hellinger risk over $\Sigma_{\text{inv}}(s, L)$ satisfies*

$$\inf_{\hat{f}} \sup_{f^* \in \Sigma_{\text{inv}}(s, L)} \mathbb{E} h^2(P^*, \hat{f}) \asymp n^{-2s/(2s+d)}, \quad d = \dim(E/G) = D - r,$$

the lower bound by Theorem 3.4 (all $s > 0$) and the upper bound by the penalised projection estimator of Theorem 5.2 (exact rate, adaptively in s). The invariant rate strictly improves on the non-invariant rate $n^{-2s/(2s+D)}$ if and only if $r > 0$.

Proof. Combine Theorems 3.4 and 5.2; the strict improvement is the strict monotonicity of $\alpha \mapsto n^{-2s/(2s+\alpha)}$ in α , with $d < D \iff r > 0$. $\square$

Remark 6.2 (Position relative to the additive literature). In the L^2 /Sobolev/kernel setting, Theorem 6.1 re-derives the dimension-reduction gain quantified by [10, 5, 3] and consistent with manifold density estimation [6, 9]. What is added is the adaptive MDL/Hellinger route, the explicit removal of the logarithmic factor (Section 5), and the multiplicative extension (Section 7).

Examples (and a negative control).

- (a) *Radial densities.* $E = \mathbb{R}^D$ or S^D , $G = SO(D)$: $d = 1$, rate $n^{-2s/(2s+1)}$ versus $n^{-2s/(2s+D)}$ — maximal gain.
- (b) *Partial translation invariance.* $E = \mathbb{T}^D$, $G = \mathbb{T}^k$: $d = D - k$.
- (c) *Finite symmetry (negative control).* Compositional data with $G = S_d$, or m -fold rotational symmetry $G = \mathbb{Z}_m$: $r = 0$, so $d = D$ and there is *no rate change*, only a constant-factor $(1/|G|)$ variance improvement. The theory thus predicts exactly when there is no gain.

7 The multiplicative / twin regime

We now take the data to be positive or compositional and the invariance to act in a logarithmic chart, where smoothness is measured. This is the regime outside the additive analyses, and it is where the tail-freeness of Lemma 4.1 pays off.

Assumption 4 (Multiplicative regime). $E \subseteq (0, \infty)^D$ (or the simplex Δ^{D-1}), with chart $\phi = \log$ (resp. the centred log-ratio), so the base operation is the twin sum $x \odot y = \phi^{-1}(\phi(x) + \phi(y))$. G acts by isometries of the twin coordinate $\phi(E)$, with orbital metric and quotient as in Assumption 1 computed in the ϕ -chart, $d = \dim(\phi(E)/G)$; smoothness $\Sigma(s, L)$ is measured in the twin coordinate, on a compact d -Ahlfors-regular region as in Assumption 2.

Assumption 5 (Chart regularity; heavy tails in the identity scale). In the ϕ -chart the transported target—the density $\tilde{f}^*$ of $\pi(\phi(X))$ on $(\phi(E)/G, \mu_G^\phi)$ with μ_G^ϕ the pushforward of the reference measure—satisfies Assumptions 2–3: it lies in $\Sigma(s, L)$ and obeys $c_0 \leq \tilde{f}^* \leq C_1$ on a compact d -Ahlfors-regular region $U_\phi \subseteq \phi(E)/G$. No moment or tail condition is imposed on X in the identity chart: X may be heavy-tailed (e.g. $\mathbb{E}[X^p] = \infty$ for some $p > 0$) provided $\phi(X) = \log X$ takes values in U_ϕ .

Theorem 7.1 (Exact rate on multiplicative data). *Under Assumptions 1, 3, 4, 5, the penalised projection estimator of Theorem 5.2 built in the ϕ -chart satisfies, for every $f^* \in \Sigma_{\text{inv}}(s, L)$,*

$$\mathbb{E}_{P^*}[h^2(P^*, \hat{f})] \asymp n^{-2s/(2s+d)}, \quad d = \dim(\phi(E)/G),$$

adaptively, and this holds on positive/compositional and heavy-tailed data for which no identity-coordinate (Hoeffding/Bernstein) concentration bound exists.

Proof. The squared Hellinger distance is a change-of-variables invariant: for the diffeomorphism ϕ , h^2 computed in the ϕ -chart with the twin-transported reference measure equals h^2 pulled back to E . Hence the lower bound (Theorem 3.4) transports verbatim, the construction being supported on a compact ball of the quotient in the ϕ -chart. For the upper bound, run Proposition 5.1/Theorem 5.2 in the ϕ -chart: the quotient entropy and the twin Jackson bound (Lemma 4.4) are metric statements that transport through ϕ , and the projection variance bound is computed against the transported reference measure. The only place tails could enter is the affinity/oracle step, which is tail-free by Lemma 4.1 and Remark 4.3; for the projection estimator the variance bound depends only on the two-sided bound $c_0/2 \leq \tilde{f}, \tilde{f}^* \leq 2C_1$ in the ϕ -chart granted by Assumption 5, not on the tails of X in the identity chart. Balancing k gives the exact rate. $\square$

Remark 7.2 (Concrete heavy-tailed instances). Assumption 5 is met, for example, by: (i) *log-normal-type* positive data, $\log X$ having a smooth density on a compact regular region; (ii) *Pareto/power-law* positive intensities, where $\log X$ has exponential (hence sub-Gaussian on bounded sets) tails while X has $\mathbb{E}[X^p] = \infty$ for p beyond the tail index; (iii) *compositional* data on the simplex in centred-log-ratio coordinates, with a continuous sub-composition symmetry. In each case the identity-chart density admits no usable Hoeffding/Bernstein constant, yet Theorem 7.1 delivers the exact rate $n^{-2s/(2s+d)}$, because the projection variance is controlled by the chart bound and the affinity oracle (Lemma 4.1) is tail-free.

Remark 7.3 (What is genuinely new here). The bare rate gain (Theorem 6.1) overlaps known results; the new content is (a) the adaptive MDL/Hellinger derivation and the exact-rate projection refinement, and (b) Theorem 7.1: the gain on multiplicative / heavy-tailed data via the canonical, scale-invariant affinity coordinate, a regime the additive invariance-gain literature does not cover. Concrete instances: positive Poisson/point-process intensities, compositional (simplex) data, and multiplicative-noise diffusions.

8 A numerical illustration

We illustrate the dimension gain and the finite-group negative control with the penalised projection estimator of Theorem 5.2 (a piecewise-constant projection, i.e. a regular histogram), reported at its risk-optimal resolution (the bias/variance balance of Proposition 5.1; the data-driven penalty attains the same rate). Targets are Lipschitz ($s = 1$), so the predicted rate in dimension d is $n^{-2/(2+d)}$. Each point is a mean integrated squared error over 30 (resp. 80) replications, fitted on a log–log scale.

(a) *Radial*, $E = \mathbb{R}^2$, $G = SO(2)$ ($d = 1$). We compare the invariant estimator (1-D projection of the radii $R_i = |X_i|$, pushed back to a radial density on $\mathbb{R}^2$) with the ambient 2-D projection, both judged as estimators of the same density f . The fitted slopes are -0.62 (invariant) versus -0.41 (ambient), to be compared with the predictions $-2/3$ and $-1/2$: the invariant estimator is clearly steeper, exhibiting the effective-dimension gain.

(b) *Finite symmetry, circle*, $G = \mathbb{Z}_4$ ($d_{\text{eff}} = D$). Here the theory predicts no rate change, only a constant variance gain. The orbit-folded and plain estimators have indistinguishable slopes (-0.59 and -0.60), while their ISE ratio is n -independent, $\approx 2.8 \leq |G| = 4$ — a pure constant-factor improvement, confirming the negative control of Theorem 6.1.

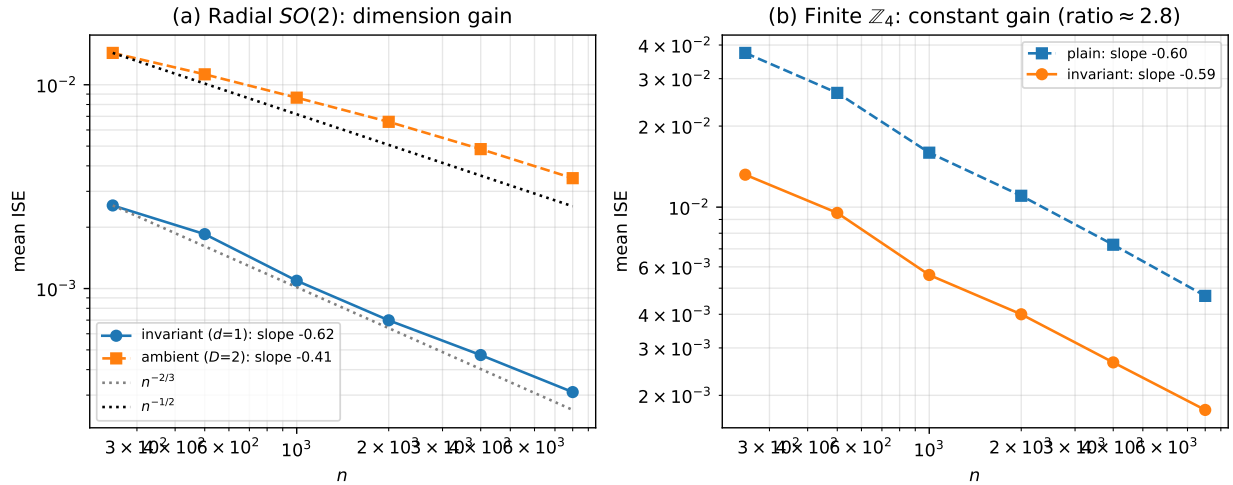


Figure 1: Penalised projection estimator. (a) Radial $SO(2)$: the invariant estimator ($d = 1$) decays at slope $-0.62 \approx -2/3$, the ambient estimator ($D = 2$) at $-0.41 \approx -1/2$ — the effective-dimension gain. (b) Finite $\mathbb{Z}_4$: equal slopes and an n -independent ISE ratio $\approx 2.8 \leq |G|$ — a constant gain only, as predicted for 0-dimensional orbits. Produced by `sim_invariant_minimax.py`.

Reproducibility. Figure 1 and all reported slopes are produced by the single script `sim_invariant_minimax.py` (Python; `numpy`, `scipy`, `matplotlib`), with a fixed seed (`numpy` generator seeded at 20260619). Running `python3 sim_invariant_minimax.py` regenerates the figure `sim_invariant_minimax.pdf` and prints, for each sample size $n \in \{250, 500, 1000, 2000, 4000, 8000\}$, the mean integrated squared errors and the fitted log–log slopes; the radial experiment averages 30 replications, the finite-group control 80. No quantity in this section is set by hand.

9 Discussion

The two scope-limiting points of the additive and multiplicative analyses are now *assumptions* under which the theorems hold, rather than gaps. The stratified case is covered by Assumption 3: when the action is not free, $M = E/G$ is an orbifold, but under Ahlfors d -regularity the singular strata are μ_G -null and the rate is governed by the top stratum M_0 . What remains genuinely open is a target whose smoothness interacts essentially with the singular locus (local dimension varying across strata), which would call for a stratified smoothness class. The multiplicative regime is covered by Assumption 5: the conclusions require only chart regularity (a compact d -regular region in the ϕ -chart with the target bounded away from 0 and ∞ there), and *no* tail condition on X , so genuinely heavy-tailed positive/compositional data are admissible (Remark 7.2). The natural next step is to relax the compact chart region to an unbounded one by controlling the integrability of $\log X$; the information-theoretic route makes this plausible because the concentration step is tail-free (Remark 4.3), so the remaining work is approximation-theoretic, on the quotient geometry, not probabilistic.

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